Representation and Communication of Digital Information II

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2026

Introduction

Communicating Technical Information

As an engineer, writing code is only half of the job.

- The other half is **communicating** what that code does.
- Audience: Other developers, managers, clients, or your future self.
- Medium: Documentation, reports, theses, or presentations.
- Objective: Use tools that allow for versioning, automation, and professional quality.

The Cost of Poor Documentation I

What happens when we don’t document?

- **Wasted Time:** Developers spending hours trying to understand a function.
- **Errors:** Misusing an API because the documentation was unclear.
- **Onboarding:** New team members taking months instead of weeks to be productive.

The Cost of Poor Documentation II

- **Maintenance:** Returning to your own code after 6 months and not knowing why you did something.
- **Technical Debt:** Unclear code/docs lead to hacks and workarounds.
- **Silos:** Knowledge is trapped in one person's head.

Markdown

What is Markdown?

A lightweight markup language with plain-text-formatting syntax.

- **Origin:** Created by John Gruber in 2004.
- **Philosophy:** “Readability, above all else.”
- **Format:** Plain text that can be converted to many formats (HTML, PDF, DOCX).
- **Usage:** GitHub (READMEs), Documentation (MkDocs, Sphinx), Academic notes.

Markdown Flavors

Markdown is not a single standard, but has “flavors”:

- **CommonMark:** The attempt to create a highly compatible standard.
- **GitHub Flavored Markdown (GFM):** Adds tables, task lists, and autolinks.
- **Pandoc’s Markdown:** The most powerful, adding citations, footnotes, and metadata.

Markdown Syntax: Headers

Use the # symbol:

- `# Title (H1)`
- `## Section (H2)`
- `### Subsection (H3)`
- `#### Sub-subsection (H4)`

Markdown Syntax: Emphasis

- `*italic*` or `_italic_`
- `**bold**` or `__bold__`
- `***bold italic***`
- `~~strikethrough~~`

Markdown Syntax: Lists I

Unordered Lists: * Item 1 * Item 2 * Sub-item 2.1 * Sub-item 2.2

Markdown Syntax: Lists II

Ordered Lists: 1. First step 2. Second step 3. Third step

Markdown Syntax: Links and Images

- **Link:** `[Link Text](URL)`
  - `[University of Aveiro](https://www.ua.pt)`
- **Image:** `![Alt Text](Path/to/image)`
  - `![Course Logo](../../assets/logo.svg)`

Markdown Syntax: Code

- **Inline:** Surround with backticks: `` `print("Hello")` ``.
- **Blocks:** Use triple backticks with language identifier.

‘‘‘python def hello(): print(“Hello World”) ‘‘‘

Markdown Syntax: Tables and Tasks (GFM)

Task	Priority	Status
Plan	High	Done
Write	Medium	Active

- **Task Lists:**

- [x] Complete Class 09
- [x] Research Class 10
- [ ] Write Class 10 slides

Diagrams in Markdown: Mermaid.js I

Many platforms (GitHub, GitLab, Obsidian) support **Mermaid** for diagrams.

‘‘‘mermaid graph TD A[Start] –> B{Is it data?} B – Yes –> C[Process] B – No –> D[End] ‘‘‘

Diagrams in Markdown: Mermaid.js II

Mermaid supports many diagram types: * **Flowcharts** * **Sequence Diagrams** * **Gantt Charts** * **Class Diagrams** * **Entity Relationship Diagrams**

It allows you to keep your diagrams **version-controlled** alongside your code.

LaTeX

What is LaTeX?

A high-quality typesetting system for technical and scientific documentation.

- **Focus:** Separating content from style.
- **Strength:** Mathematical notation, citations, and large documents.
- **Engine:** Based on TeX, created by Donald Knuth.
- **Standard:** Used for MSc Theses, PhD Dissertations, and Scientific Papers.

LaTeX Basics: Structure

```latex
\documentclass{article}
\usepackage[utf8]{inputenc}

\title{My Report}
\author{Mário Antunes}

\begin{document}
\maketitle

\section{Introduction}
Hello world!
\end{document}
```

LaTeX Basics: Document Classes

The `\documentclass{...}` defines the overall layout:

- **article:** For short reports or papers.
- **report:** For longer documents with chapters (like an MSc Thesis).
- **book:** For full-length books.
- **beamer:** For creating presentation slides.

Beamer: Presentations in LaTeX

Beamer allows you to create PDF slides using LaTeX syntax.

- **Frames:** Each slide is a `\begin{frame} ... \end{frame}`.
- **Themes:** Professional, academic looks (e.g., `Warsaw`, `Madrid`).
- **Overlays:** Control when content appears (e.g., `\pause`).
- **Consistency:** The math and code in your slides will look identical to your paper/thesis.

LaTeX Basics: Packages

Packages extend LaTeX’s capabilities:

- **graphicx:** For including images (`\includegraphics`).
- **hyperref:** For clickable links and cross-references.
- **geometry:** For changing page margins.
- **amsmath:** For advanced mathematical symbols.
- **biblatex:** For managing bibliographies.

LaTeX Basics: Mathematics I

LaTeX is the industry standard for writing math.

- **Inline:** $E = mc^2$.
- **Display:**

$$\int_a^b f(x)dx$$

LaTeX Basics: Mathematics II

- **Fractions:** `\frac{numerator}{denominator}`
- **Sums:** `\sum_{i=0}^{n} x_i`
- **Greek Letters:** `\alpha, \beta, \gamma, \pi`
- **Sub/Superscripts:** `x_i, x^2`
- **Matrices:** `\begin{pmatrix} a & b \\ c & d \end{pmatrix}`

Citations with BibTeX/BibLaTeX

Managing references manually is a nightmare.

1. Create a `.bib` file with your references.

   ```
   @article{einstein1905,
     author = {Albert Einstein},
     title = {On the Electrodynamics of Moving Bodies},
     journal = {Annalen der Physik},
     year = {1905}
   }
   ```

2. Cite it: `Einstein showed that \cite{einstein1905}...`

Pandoc

The “Swiss-army knife” of Documents

Pandoc is a command-line tool to convert files from one markup format into another.

- **Input:** Markdown, LaTeX, HTML, DOCX, EPUB.
- **Output:** PDF, HTML, LaTeX, Slides (Beamer/RevealJS), DOCX.
- **Command:** `pandoc input.md -o output.pdf`

How Pandoc Works I

1. **Reader:** Parses the input (e.g., Markdown).
2. **Intermediate Representation:** Converts to an internal "AST" (Abstract Syntax Tree).
3. **Writer:** Generates the output (e.g., PDF via LaTeX).

How Pandoc Works II: Metadata

- **YAML Metadata:** Use a `config.yml` or a block at the top of the `.md`.
- It controls titles, authors, dates, and template variables.
- **Templates:** You can provide your own LaTeX or HTML template to control every detail of the output.

Pandoc Filters

Pandoc's power can be extended using **filters**.

- Filters are small scripts that modify the document's internal AST before it is written.
- **Types:** Python filters (`panflute`), Lua filters (built-in).
- **Usage:**
  - Automatically number figures.
  - Transform table formats.
  - Inject custom LaTeX for specific blocks.

Useful Documents

The Modern CV

- **Format:** Keep it simple, searchable, and professional.
- **Tools:** Use LaTeX templates (e.g., `moderncv`) or Markdown + Pandoc.
- **Key Sections:** Education, Experience, Skills, Projects.
- **Pro Tip:** Keep your CV in a Git repository and generate multiple versions (e.g., detailed vs 1-page).

The MSc Thesis: Typical Structure

1. **Front Matter:** Title, Abstract, Acknowledgments, ToC.
2. **Introduction:** Motivation, Problem, Objectives, Contributions.
3. **State of the Art:** Literature review and comparison.
4. **Proposed Solution/Methodology:** Architecture, design, and implementation.
5. **Evaluation:** Experimental setup, results, and discussion.
6. **Conclusion:** Summary and Future Work.
7. **Back Matter:** Bibliography, Appendices.

MSc Thesis: The “Contribution”

The most important part of your thesis is your original contribution.

- It’s not just “building a system”.
- It’s “solving a problem” or “improving a process” using engineering principles.
- You must clearly state **what is new** in your work.

Project Reports

- **Executive Summary:** For busy managers.
- **Methodology:** How you did it.
- **Findings:** What you discovered.
- **Recommendations:** What should be done next.
- **Automation:** Use a `Makefile` to generate reports from Markdown.

Python Documentation I: Tools

- **Sphinx:** The industry standard for Python docs.
- **MkDocs:** Modern, fast, and uses Markdown.
- **Docstrings:** Write documentation inside your code.

Python Documentation II: Docstrings

```python
def calculate_area(radius: float) -> float:
    """Calculates the area of a circle.

    Args:
        radius: The radius of the circle (must be positive).

    Returns:
        The calculated area.

    Raises:
        ValueError: If radius is negative.
    """
    if radius < 0:
        raise ValueError("Radius cannot be negative")
    return 3.14159 * (radius ** 2)
```

Summary

Summary

- **Communication:** Is a core engineering skill.
- **Markdown:** For daily notes, READMEs, and quick reports.
- **LaTeX:** For formal academic and high-quality technical documents.
- **Pandoc:** To bridge all formats and automate your workflow.
- **Consistency:** Use professional templates and version control for everything.